Ajay Venkatesh

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Skills

Programming Languages: C++, Java, Python, TypeScript, JavaScript (ES6), SQL
Web Technologies: HTML, CSS, React, REST API, GraphQL API, Microservices, Node.js, Express.js, Sequelize
Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse
Cloud: AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs)
Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Kubernetes, Power BI, QuickSight
AI/ML: LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, RL.

Experience

Campus Mesh — Jan 2026 – Present

Software Engineer | *Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines*

- Developed and maintained scalable **Node.js microservices** powering real-time academic workflows for **10K+ users**, including study groups, messaging, notifications, and profile management.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing user engagement by **28%** and retention by **50%**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

Amazon — May 2025 – Aug 2025

Software Development Engineer Intern | *Serverless AWS infrastructure, React 18 frontend, and Java backend systems*

- Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the **system design** for approval and established Role Based Access **(4+ roles)** using **OAuth, Midway SSO, and HELIS**.
- Provisioned **serverless Infrastructure-as-Code** using **AWS CDK** with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by **70%** and enabling **multi-stage, multi-region CI/CD deployments.**
- Built a **React 18** frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by **40%**.
- Engineered backend services in **Java** using **Coral, Smithy, Dagger**, and **CBOR serialization**; **applied low-level design and design patterns** reduced backend execution time by **35%** and ensure reliability by unit and integration testing using **Mockito**.

New York University (Novel AI Technologies, Inc.) — Aug 2024 – Present

Software Developer (On-Campus) | *NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics*

- Architected an **NSF-funded ($100K)** AI-powered health app for **iOS** and **Android**, processing real-time wearable data; deployed cloud-hosted AI models for health metrix analysis and process food images.
- Engineered ETL pipelines transforming **NoSQL** into **PostgreSQL** using **concurrency control and locking mechanisms**; processed **250K+** records and powered **AWS QuickSight** dashboards for real-time health analytics.
- Developed an insurance analytics platform **(React, Node.js, REST APIs)**; deployed **(Docker** containers on **EC2**, implemented Stripe billing, RBAC, and row-level security, generating **$50K+** revenue.
- Streamlined underwriting via broker portal integrated with insurer platform; implemented **document and media processing** and **WebSocket-based real-time communication**; secured **6 insurance clients**.

PricewaterhouseCoopers (PwC) — Aug 2021 – Aug 2024

Senior Software Engineer | *Cloud-native backend, NLP/OCR automation, and data engineering on Azure*

- Developed a distributed backend on **Microsoft Azure** for a health platform, integrating with **Microsoft CRM** and implementing event-driven processing and fault tolerance.
- Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.
- Deployed an **NLP** based **OCR neural model** on Azure to automate structured data extraction, reducing data processing time by **75%**.
- Optimized data ingestion with **parallel processing** for **100K**-row from Excel; reduced execution time from **5 hours to 1 hour**.
- Built **Power BI dashboards** processing large datasets from multiple databases, improving reporting efficiency by **40%**.

Projects

Advanced Expense Tracker (Python Desktop Application) | ***Python, Tkinter, SQLite3, Pandas, Matplotlib***

GUI-based personal finance manager with real-time data visualization and concurrent multi-user transaction handling

- Architected a modular **Python application** supporting individual and group expenses with secure login, applying **OOP principles** and **MVC & DAO design patterns** with **SQLite3** for persistent storage.
- Implemented concurrent transaction handling and real-time cost splitting using **Python threading** and locks to ensure thread-safe database writes; visualized spending trends with **Matplotlib** charts.

Hospital Management System (Full-Stack Application) | ***React, Node.js, Express, MongoDB, SageMaker, Hugging Face***

- Built a **full-stack healthcare platform** using **React, Node.js, Express, and MongoDB** to manage appointments, patient records, lab reports, and medical history with **role-based access control**.
- Fine-tuned a **Hugging Face transformer-based model** achieving **88% accuracy** for health condition prediction and deployed it on **AWS SageMaker** for real-time diagnostic support in the doctor dashboard.

Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** — Sep 2024 – May 2026

- *Coursework:* System Design, Machine Learning, Deep Learning, Big Data, Advanced Python, Data Structures

Certifications

AWS Cloud Practitioner
Azure Data Fundamentals
Microsoft Solution Architect Expert
Power BI Data Analyst Associate
Microsoft Developer Associate
Stanford Machine Learning (Coursera)